

Calibration-Aware Personalized Federated Learning for Molecular Reorganization Energy Prediction

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Abstract

Molecular reorganization energy is a primary descriptor in the design of organic semiconductors and thermally activated delayed fluorescence (TADF) emitters, yet a single experimental laboratory typically produces only a few dozen high-fidelity samples — far below what deep models require for robust generalization. Federated learning offers a path to pool laboratories and public datasets without exchanging raw structures, but the prevailing toolkit (FedAvg, FedProx, FedPer, FedBN, MOON) was designed for statistical heterogeneity, not for the systematic heterogeneity that distinguishes quantum-chemical regression: distinct physical quantities (hole- λ versus triplet- λ), distinct DFT protocols (B3LYP/6-31G* versus ω B97X-D with adiabatic four-point Nelsen), and label scales that can differ by more than a factor of two.

We introduce **PC²-FedReorg** (Protocol-Calibrated Personalized Federated Learning), a federation that distinguishes *physical clients* from *task-clients*: distinct quantities measured at the same laboratory are modeled as distinct task-client nodes, each with a private (μ, σ) calibration buffer that is computed strictly from its own training fold and universally excluded from cross-client aggregation. Encoders and an optional residual adapter are shared under a per-key routing scheme defined by an auditable chemistry/protocol transferability matrix; regression heads and calibration buffers remain private.

On a five-node federation (A: QM9 non-aromatic, $n = 6,020$; B: QM9 aromatic, $n = 9,190$; C-hole: in-house TADF, $n = 53$; C-triplet: in-house TADF, $n = 49$; D: Atahan-Evrenk 2019, $n = 5,876$) evaluated by five-seed leave-one-out cross-validation, PC²-FedReorg achieves **nominally significant** MAE reductions on triplet reorganization energy compared with FedAvg, FedProx and FedPer (paired Wilcoxon $p = 0.045$, 0.045 and 0.042 ; MAE 0.66 vs. 0.76–0.78 eV), while no statistically separable improvement is observed on hole reorganization energy, consistent with the small-sample regime already evident in classical baselines. A targeted control (E71) further shows that a FedPer baseline endowed only with task-specific calibration—without the residual adapter and without the transferability-driven routing—yields a C-triplet MAE that is not statistically distinguishable from PC²-FedReorg’s at our sample size (per-molecule median-aggregated MAE 0.654 vs. 0.660 eV; paired Wilcoxon $p = 0.929$; bootstrap 95% CI on Δ MAE straddles zero), while yielding nominally lower MAEs than FedPer without calibration (paired Wilcoxon $p = 0.028$) and than the no-calibration ablation of PC²-FedReorg (paired Wilcoxon $p = 0.021$). The corresponding 5-seed mean MAEs follow the same ordering (0.653 ± 0.011 vs. 0.671 ± 0.010 eV for E71 vs. PC²-FedReorg). These results are consistent with task-specific calibration under personalized heads carrying the only ablation-detectable signal in our federation; we did not pre-register an equivalence margin and therefore do not interpret the n.s. comparison as evidence that the adapter and the chemistry/protocol gate contribute nothing, only that any additional benefit they confer is below detection at

our seed budget and sample size. The transferability matrix is retained as a deterministic, inspectable record of which source–target sharings the algorithm permits—persisted before any training begins—independently of the optimization outcome.

Keywords: personalized federated learning; molecular reorganization energy; task-specific calibration; label-scale heterogeneity; quantum chemistry; organic semiconductors; TADF emitters; small-sample molecular learning.

1 Introduction

The Marcus picture of charge-transfer kinetics identifies the molecular reorganization energy λ — the energetic cost of relaxing the equilibrium geometry following an electron- or hole-transfer event — as a primary determinant of charge mobility in organic field-effect transistors, hole/electron transport layers in organic light-emitting diodes (OLEDs), and the radiative–nonradiative balance in TADF emitters [1–4]. Reliable *in silico* λ -screening accelerates the experimental design loop and reduces wet-lab attrition.

In practice, however, λ prediction sits at an uncomfortable intersection. High-fidelity reference values require multi-point geometry optimization and DFT single-point evaluation at consistent functional/basis combinations; a single experimental TADF laboratory typically generates 50–100 such labels per year. Conjugated-molecule λ predictions from neural networks have therefore relied on aggregating laboratories’ data into a single pool [5, 6], at the cost of losing protocol consistency and obscuring the resulting label-scale heterogeneity.

Federated learning (FL) [7] is a natural response to this asymmetry: a private TADF laboratory can contribute gradient updates to a shared model without releasing its raw molecules, while industrial precompetitive consortia and public-data hosts can participate symmetrically. The standard FL tool-kit — FedAvg [7], FedProx [8], FedPer [9], FedBN [10], and contrastive variants such as MOON [11] — was developed for statistical heterogeneity across clients drawn from the same underlying distribution, not for the systematic heterogeneities present in quantum-chemical labels: (i) the physical quantity being regressed (hole- λ vs. triplet- λ) may differ across clients hosted at the same laboratory; (ii) DFT protocols are unique fingerprints (functional, basis, geometry treatment, charge state, conformer policy), and protocol-mismatched labels are not interchangeable; (iii) label scales can differ by $2\times$ or more — the same head trained with mean-squared-error loss on $z = 0.26$ eV (D) and $z = 2.64$ eV (C-triplet) will optimise primarily toward the larger-scale client, regardless of its protocol compatibility.

A natural remedy is to model these heterogeneities explicitly. We propose **PC²-FedReorg**, a personalized federation that (i) distinguishes physical clients from *task-clients* defined by quantity/protocol/scale; (ii) attaches a private, fold-isolated (μ, σ) calibration buffer to each task-client so that loss and metric live in compatible spaces; (iii) maintains a layer-wise separation between universally shared encoder/adaptor parameters and quantity-specific private head/calibration parameters; and (iv) encodes the chemistry- and protocol-derived compatibility information into an auditable 5×5 transferability matrix that documents which source–target sharings are permitted at aggregation. Section 4.4 reports a targeted control (E71) showing that, in our federation, the operative element is component (ii) under personalized heads; components (iii) and (iv) serve as auditability scaffolding rather than additional performance drivers, and the framework is therefore

better described as *calibration-aware personalized federated learning made auditable*.

PC²-FedReorg has five design components. We list them in decreasing order of empirical support in this study, with explicit boundaries on what the present ablations can and cannot conclude:

1. **Task-specific calibration under personalized heads as the only ablation-detectable component.** A non-trainable (μ, σ) buffer per task-client, computed strictly from each LOOCV training fold and universally excluded from every federated aggregation operator, paired with a private regression head. Loss is computed in z-space; metrics are reported in eV-space. A targeted control (§4.4, E71) shows that a FedPer baseline endowed only with this buffer — without the residual adapter and without the transferability-driven routing — yields a C-triplet MAE that is not statistically distinguishable from PC²-FedReorg’s at our sample size (paired Wilcoxon $p = 0.929$; 95 % bootstrap CI on ΔMAE straddles zero), while yielding nominally lower MAEs than FedPer without calibration ($p = 0.028$) and than the no-calibration ablation of PC²-FedReorg ($p = 0.021$). Calibration under a private head is therefore the only component whose presence has measurable support in our ablation set; we did not pre-register an equivalence margin and therefore do not claim that the adapter or the gate contribute nothing, only that any additional benefit they confer is below detection at our sample size.
2. **Task-client modeling for heterogeneous cation/hole/triplet λ tasks.** A federated formulation in which hole- λ , triplet- λ and cation- λ measurements are treated as distinct task-client nodes throughout aggregation, head parameters and calibration, even when two task-clients are hosted by the same physical laboratory.
3. **A calibration-aware personalized federated baseline that reproduces PC²-FedReorg gains.** The control variant in (1) — FedPer with a per-task-client (μ, σ) buffer, no residual adapter, no transferability matrix consulted at aggregation — serves as a transparent baseline for heterogeneous-quantity federations and recovers the headline C-triplet improvement of the full PC²-FedReorg architecture.
4. **Deterministic, inspectable compatibility metadata for protocol/quantity constraints.** A 5×5 transferability matrix derived from Morgan-Tanimoto similarity, Murcko scaffold overlap and conservative DFT-protocol clauses, providing a training-data-independent record of which source–target pairs are *permitted* to share parameters. The matrix is computed before any aggregation round, persisted to `T_transferability.json` with the failing-clause `reasons` field, and can be inspected without any model state. We claim three concrete properties for this artifact: it is *deterministic* (a fixed function of declared metadata), *inspectable* (a single JSON file rather than a model-state dump), and *aggregation-operator-independent* (the same exclusions apply under FedAvg, FedProx, FedPer, FedBN and PC²-FedReorg through a shared `_get_exclude_keys` filter). We do **not** claim that the matrix is a performance lever — the ablations and the E71 control in §4.4 do not detect a performance contribution from the chemistry-driven gate weighting over uniform weighting in our particular federation at the present sample size — and we do not test downstream compliance-violation or contamination-prevention scenarios in this manuscript.

5. **A reliable small-sample evaluation protocol.** Five-seed LOOCV with paired Wilcoxon signed-rank tests and bootstrap 95 % confidence intervals on the paired MAE difference; MAE / RMSE are primary, R^2 is auxiliary.

Section 2 specifies the federation; Section 3 presents the datasets; Section 4 reports the main empirical findings; Section 5 discusses the active ingredient and the role of the gate; Section 6 lists the limitations of the present study; Section 7 concludes.

2 Methods

2.1 Task-client formulation

Let an organic-electronics federation consist of five task-client nodes (Figure 1; Table 3):

Table 1: Five task-client nodes used in this study (see also Table 3 for full per-node statistics).

Node	Source	Physical quantity	DFT protocol summary
A	QM9 non-aromatic public	cation reorg. λ	same public cation-reorg source protocol; head-sharing with B permitted by same-source shortcut (§ 2.4)
B	QM9 aromatic public	cation reorg. λ	same public cation-reorg source protocol; head-sharing with A permitted by same-source shortcut (§ 2.4)
C-hole	in-house TADF lab	hole- λ (neutral $\rightarrow$ cation)	ω B97X-D, adiabatic 4-point Nelsen [12]
C-triplet	in-house TADF lab	triplet- λ (neutral $\rightarrow$ T ₁)	ω B97X-D, adiabatic 4-point Nelsen [12]
D	Atahan-Evrenk 2019 [5]	hole- λ	B3LYP/6-31G* reported; geometry/charge-state/conformer policy not stated in source metadata

Critically, C-hole and C-triplet are different *task-clients* even though their underlying molecular structures largely overlap. They measure different physical quantities (different end states), and their label distributions differ by a factor of 2.17 (means 1.215 vs. 2.638 eV; Figure 1b). Merging them into a single regression task corrupts the head’s quantity-specific mapping.

The task-client abstraction is a modeling assumption. It does not change which DFT logs are read or how molecules are graphed; it changes how parameters are routed and where (μ, σ) buffers are installed.

2.2 Model architecture

Each task-client owns a model decomposed into four logical stages (Figure 2):

```

Mol  $\rightarrow$  Encoder (5-layer GIN, projected to 256-d) [shared, T_repr-gated]
       $\rightarrow$  Adapter (residual 256 $\rightarrow$ 128 $\rightarrow$ 256, zero-initialised to identity)
                                         [optionally shared, T_adapter-gated]
       $\rightarrow$  Head (KAN or MLP, quantity-specific) [PRIVATE]
       $\rightarrow$  Calibration (frozen  $(\mu, \sigma)$  buffer) [PRIVATE]
       $\rightarrow$   $\hat{\lambda}$  (eV)

```

- The encoder produces a 256-d chemical-representation vector; we use a five-layer GIN [13] with Jumping-Knowledge concatenation and mean+max pooling, consistent with prior molecular FL benchmarks. Our codebase supports both GIN and SchNet encoders through a common interface; all reported PC² and baseline runs in this manuscript use the 5-layer GIN encoder for fair comparison.
- The adapter is a residual bottleneck initialised to the identity so that, prior to training, removing it has no effect on the network output.
- The head is a three-layer KAN [14] (`efficient-kan` implementation [15]) or three-layer MLP; head parameters are *never* shared across task-clients.
- The calibration carries non-trainable buffers (μ_c, σ_c) per task-client c , installed from the training fold’s labels. Training loss is computed in z-space; predictions are de-standardised at inference (Figure 2, formula band).

2.3 Federated training and per-key aggregation

Training alternates between local supervised epochs and a round of aggregation. We define aggregation as a per-state-dict-key operation routed by layer kind:

Table 2: Per-key aggregation routing under PC²-FedReorg. The same exclusion filter applies under every federated strategy benchmarked in this study (FedAvg/FedProx/FedPer/FedBN/PC²).

Key substring	Weighting	Source
<code>encoder.*</code>	$T_{\text{repr}} \cdot n_i$, renormalised	shared with chemistry gate
<code>adapter.*</code>	$T_{\text{adapter}} \cdot n_i$, renormalised	shared with geom. mean gate
<code>head.*</code>	$T_{\text{head}} \cdot n_i$, renormalised	quantity-compatible sources only
<code>calibration.*</code>	skipped	never aggregated
<code>bn.*, *.norm.*</code>	skipped	FedBN-style local statistics [10]

Aggregation is **per-target**: for each target task-client j and each shared parameter key belonging to layer kind $\ell \in \{\text{encoder}, \text{adapter}, \text{head}\}$, the contribution of source task-client i is weighted by

$$w_{i \rightarrow j}^{(\ell)} = \frac{T_{ij}^{(\ell)} n_i}{\sum_k T_{kj}^{(\ell)} n_k}, \quad (1)$$

where n_i is the supervised sample count of task-client i and $T^{(\ell)}$ is the corresponding transferability matrix (Section 2.4). When all $T_{kj}^{(\ell)}$ equal zero for a given target j , the target keeps its local copy of the layer (an identity routing fallback that prevents division-by-zero and preserves the FedPer “private when nothing is compatible” intuition). PC²-FedReorg’s universal exclusion of calibration is enforced through a shared `_get_exclude_keys` filter applied under every federated strategy (FedAvg / FedProx / FedPer / FedBN / PC²), so that strategy-specific differences are isolated to the aggregation operator itself.

2.4 Auditable transferability matrices

Three 5×5 transferability matrices are pre-computed once before training and persisted as `T_transferability.json`. They are deterministic functions of the chemistry and protocol

metadata — *independent of any label or model state* — and can be inspected and audited prior to any aggregation round.

T_{repr} (**encoder gate, chemistry-driven**):

$$T_{\text{repr}}[i, j] = \max(\epsilon_r, w_K K_{ij} + w_S S_{ij}),$$

with K_{ij} the mean over i -molecules of max-Tanimoto-similarity to j -molecules (Morgan radius 2, 2048 bits; computed with RDKit [16]), S_{ij} the asymmetric Murcko scaffold overlap, $(w_K, w_S) = (0.6, 0.4)$, and $\epsilon_r = 10^{-3}$.

T_{head} (**head gate, conservative protocol clauses, evaluated in this order**):

1. **Clause 1 – quantity mismatch:** if $\text{quantity}(i) \neq \text{quantity}(j)$, set $T_{\text{head}}[i, j] = 0$.
2. **Clause 4 – V1 hard rule:** if $i = \text{D}$ and $j \in \{\text{C-hole}, \text{C-triplet}\}$, set $T_{\text{head}}[i, j] = 0$ (applied early so the JSON `reasons` field always attributes $\text{D} \rightarrow \text{C}^*$ to the V1 evidence rather than to any later metadata clause).
3. **Same-source shortcut:** if quantity matches *and* $\text{source_id}(i) = \text{source_id}(j)$, the pair is treated as protocol-compatible by source and the per-field checks (clauses 2 and 3 below) are skipped. This shortcut models the fact that public datasets distributed from a single source are computed by a single canonical protocol, which need not be re-verified field-by-field. In our federation, the shortcut fires only for $\text{A} \leftrightarrow \text{B}$ (both derived from the QM9-public split).
4. **Clause 2 – unknown protocol field:** if the shortcut did not fire and *any* of `{functional, basis, geometry, charge_state, conformers}` equals "unknown" on either side, set $T_{\text{head}}[i, j] = 0$.
5. **Clause 3 – protocol-field mismatch:** if the shortcut did not fire and any populated protocol field disagrees between i and j , set $T_{\text{head}}[i, j] = 0$.

For pairs that survive the above, $T_{\text{head}}[i, j]$ equals a label-distribution similarity term

$$L_{ij} = \exp(-\alpha_W W_1(\hat{p}_i^{\text{label}}, \hat{p}_j^{\text{label}})) \cdot \min(\sigma_i/\sigma_j, \sigma_j/\sigma_i)^{\gamma_L},$$

with $\alpha_W = 5.0$ and $\gamma_L = 1.0$; values below 0.01 are hard-floored to zero. The clause order above matches the deterministic order in `src/transferability.py:t_head_compatibility`, so the `reasons` field of `T_transferability.json` always reports the first failing clause; in our federation this is clause 1 for $\text{A/B} \rightarrow \{\text{D}, \text{C-hole}, \text{C-triplet}\}$, clause 4 for $\text{D} \rightarrow \text{C-hole}$, and clause 1 for $\text{D} \rightarrow \text{C-triplet}$.

T_{adapter} (**interpolating gate**):

$$T_{\text{adapter}}[i, j] = \max\left(\epsilon_a, \sqrt{T_{\text{repr}}[i, j] \cdot T_{\text{head}}[i, j]}\right).$$

These rules generate several invariants automatically (Figure 3, T_{head} matrix):

- $T_{\text{head}}[\text{D}, \text{C-hole}] = 0$ (clauses 2 + 4),
- $T_{\text{head}}[\text{D}, \text{C-triplet}] = 0$ (clause 1),
- $T_{\text{head}}[\text{C-hole}, \text{C-triplet}] = T_{\text{head}}[\text{C-triplet}, \text{C-hole}] = 0$ (clause 1),
- $T_{\text{head}}[\text{A}, \text{C-triplet}] = T_{\text{head}}[\text{B}, \text{C-triplet}] = 0$ (clause 1).

We emphasize that this matrix is intended as **auditable algorithmic transparency**: it documents which source–target sharings are permitted, independently of any optimization outcome. Whether the gate also improves performance over uniform weighting is a separate empirical question, examined in Section 4.4.

2.5 Evaluation protocol

For Clients A, B and D we use 5-fold cross-validation. For C-hole ($n = 53$) and C-triplet ($n = 49$) we use leave-one-out cross-validation (LOOCV). Both protocols are repeated for 5 random seeds {42, 123, 456, 789, 1000}; the per-molecule LOOCV ordering is deterministic.

Per-molecule absolute errors are aggregated as the median across the five seeds, yielding a single 53- or 49-element per-method error vector for paired analysis. Method pairs are compared by:

1. Paired Wilcoxon signed-rank test (two-sided, `zero_method='wilcox'`) on the per-molecule absolute errors.
2. Bootstrap 95 % confidence interval on the paired MAE difference (5,000 resamples of molecule indices).
3. Win / loss / tie counts per molecule.

MAE and RMSE are reported as primary metrics. R^2 is reported only as an auxiliary; for $n \leq 53$ LOOCV regimes, classical baselines (RF, KNN, GP) already produced negative R^2 (best classical $R^2 = -0.10$ on C-hole, $+0.08$ on C-triplet), which we interpret as a small-sample data ceiling rather than a methodological flaw.

3 Datasets and Experimental Setup

3.1 Datasets

Table 3 summarizes the five task-client nodes. Total label volume is approximately 21,180 molecules, but only ~ 100 reside in the two private TADF nodes that form the small-sample target of this study (Figure 1).

Public Clients A and B are derived from a single public QM9 cation-reorganization-energy split (column `reorg_energy_eV` of `public_reorg_energy_15210.csv` in the `client_a_b` data directory), partitioned by aromatic-ring count (`NumAromaticRings == 0` for A, ≥ 1 for B). Both clients are assigned the same `source_id` in the implementation; their pairwise A $\leftrightarrow$ B head-sharing is permitted via the same-source shortcut documented in Section 2.4. Client D (Atahan-Evrenk 2019 [5]) measures hole reorganization energy on a conjugated-organic-semiconductor library; the

published metadata report B3LYP/6-31G* but geometry treatment, charge-state handling, and conformer policy are not stated.

The deterministic outcomes of the head gate for D and the C-pair are:

- **D $\rightarrow$ C-hole:** clause 4 (V1 hard rule) fires and is reported first; clauses 2 (protocol fields unknown for D) and 3 (functional family differs: B3LYP vs. ω B97X-D) would each also block this pair independently. We therefore treat $T_{\text{head}}[\text{D}, \text{C-hole}] = 0$ as a multiply-determined invariant, not contingent on any single piece of evidence.
- **D $\rightarrow$ C-triplet:** clause 1 (quantity mismatch: hole- λ vs. triplet- λ) is sufficient on its own; the metadata gap for D is irrelevant here.

These outcomes can be audited directly from the `reasons` field of the released JSON artefact `T_transferability.json` under `results/preverify/` (Figure 3).

The in-house TADF laboratory provides Clients C-hole ($n = 53$ molecules with valid hole- λ at ω B97X-D) and C-triplet ($n = 49$ molecules with valid triplet- λ at ω B97X-D). The two C clients share the majority of underlying molecular scaffolds (Figure 3, T_{repr}) but measure different end-state geometries and therefore yield label means differing by $2.17\times$ (Figure 1, label-scale heterogeneity panel).

3.2 Training setup

Encoders are five-layer GIN networks (hidden dim 300, JK-concat, mean+max pooling, projection to 256-d). All KAN heads use `grid = 5` and `spline order = 3`, following the `efficient-kan` configuration [15] used throughout this codebase. Adapters are 256-128-256 residual bottlenecks (zero-initialised). Optimiser Adam, learning rate 1×10^{-3} for local epochs, cosine annealing. Per-round local epochs = 5 for public clients, 10 for private clients; communication rounds = 30. Batch sizes $B = 512$ (A, B, D) and full-batch (C). Training is performed on dual NVIDIA A30 (24 GB each) with Client A and D placed on `cuda:0` and Clients B, C-hole, C-triplet placed on `cuda:1`; aggregation is performed on CPU.

The same code path runs all federated strategies (FedAvg / FedProx / FedPer / FedBN / PC²) through a common `get_aggregation_fn` dispatch and `_get_exclude_keys` filter, so that strategy-specific differences are isolated to the aggregation operator and the per-key routing logic.

4 Results

4.1 Task-client heterogeneity along quantity, protocol, and label-scale axes

Figure 1 visualises the five task-client nodes with their representative molecules — each selected deterministically as the molecule whose label is closest to the per-node median. The accompanying bar in Figure 1a quantifies label-scale heterogeneity: D-hole sits at 0.26 eV (mean), A and B at 0.66–0.82 eV, C-hole at 1.22 eV, and C-triplet at 2.64 eV — a $10\times$ spread. Figure 1b casts all five label distributions onto a common axis, demonstrating that no two task-clients share a comparable label scale.

Table 3 lists the per-node DFT protocols. The five task-clients fall into three distinct quantity classes: A and B are cation-reorganization-energy task-clients (public QM9-derived); C-hole and

D are hole-reorganization-energy task-clients but they differ in functional (ω B97X-D vs. B3LYP), in published protocol fields (full Nelsen four-point treatment vs. abstract-only metadata), and in chemical space; C-triplet is a triplet-reorganization-energy task-client. All five quantities are reorganization-energy-related but they are not interchangeable regression targets. The conservative T_{head} clauses (Section 2.4) detect each form of incompatibility: clause 1 (quantity mismatch) blocks every A/B/D pair against C-triplet; clause 1 also blocks A/B against the hole- λ pair (D, C-hole) in the current code, which labels the public QM9 split with a generic “reorg-energy” quantity tag rather than a specific cation label to keep public-private head sharing conservative; clauses 2–4 block $D \rightarrow \text{C-hole}$ jointly (Section 2.4).

These observations motivate the modeling decision that the same lambda symbol does not refer to the same quantity across our five nodes, and that label scales prevent a naive shared regression head from being well-posed. Chemical-space heterogeneity is examined separately through the transferability matrices in Section 4.2.

4.2 Auditable compatibility metadata generates non-trivial constraints from evidence

Figure 3 plots the three transferability matrices. T_{repr} shows that the C-hole and C-triplet molecule sets are essentially identical from the chemistry perspective ($T_{\text{repr}} \approx 0.98\text{--}1.00$, asymmetric under our $K + S$ definition), while D is moderately distant from the C nodes ($T_{\text{repr}} \approx 0.05$) and the A/B public sets sit in between. T_{adapter} , as the geometric mean, inherits the same general topology.

T_{head} shows a deliberately sparser pattern: only $A \leftrightarrow B$ receives a non-zero off-diagonal (0.33), reflecting their shared quantity and DFT protocol. All other off-diagonals are zero. Importantly, the four invariants $T_{\text{head}}[\text{D}, \text{C-hole}] = 0$, $T_{\text{head}}[\text{D}, \text{C-triplet}] = 0$, $T_{\text{head}}[\text{C-hole}, \text{C-triplet}] = 0$, and $T_{\text{head}}[\text{C-triplet}, \text{C-hole}] = 0$ (highlighted in Figure 3) follow deterministically from the protocol metadata and quantity definitions; they do not depend on any training-time signal and can be audited from `T_transferability.json` prior to any aggregation round.

We emphasize the framing: the chemistry/protocol gate is presented here as an algorithmic transparency mechanism, *not* as a claim that the gate per se improves performance over uniform weighting. The empirical decomposition is the subject of Section 4.4.

4.3 PC²-FedReorg achieves nominally significant gains on the triplet task; the hole task is at a data ceiling

Figure 4 reports the main performance comparison on the two private TADF targets, with full numerical breakdown in Tables 4 and 5.

On C-triplet ($n = 49$ LOOCV) PC²-FedReorg achieves MAE = 0.660 eV, lower than FedAvg (0.763 eV), FedProx (0.784 eV) and FedPer (0.762 eV) by approximately 13–16 % relative. Paired Wilcoxon signed-rank tests on per-molecule absolute errors give $p = 0.045$ vs. FedAvg, $p = 0.045$ vs. FedProx, and $p = 0.042$ vs. FedPer (Table 5; Figure 4 right). We describe these as **nominally significant** rather than significant: the p -values are unadjusted, sit close to the conventional 0.05 threshold, and would *not* survive a Bonferroni correction for the three baseline comparisons performed in this section ($\alpha/3 \approx 0.017$). The bootstrap 95 % confidence interval on the PC²-vs-

FedAvg paired MAE difference is $(-0.220, +0.004)$ — the upper bound only just touches zero. For the PC^2 -vs-FedProx and PC^2 -vs-FedPer comparisons the CI upper bound sits just below zero $((-0.254, -0.007)$ and $(-0.205, -0.002)$ respectively); both upper bounds are within ~ 0.01 eV of zero and shift across the zero line under a different bootstrap RNG seed (independent reseed in `paper/audit/` yields $(-0.259, +0.002)$ and $(-0.209, +0.007)$ respectively), so the CI-exclusion-of-zero criterion is fragile at these effect sizes and the Wilcoxon p -values remain the primary inferential anchor. Per-molecule win/loss counts (PC^2 better / worse) are 31 / 18, 30 / 19 and 32 / 17 respectively (Table 5). We therefore interpret the triplet results as a small-sample but reproducible signal rather than a broad performance guarantee.

On C-hole ($n = 53$ LOOCV) All evaluated methods produce MAE in a narrow band of 0.368–0.414 eV (Figure 4 left, Table 4). The eight paired comparisons we performed all return $p \geq 0.20$ after rounding (Table 5); the bootstrap CIs straddle zero in every case. PC^2 -FedReorg is statistically indistinguishable from FedAvg, FedProx, FedPer or local-only training on this target.

We attribute the C-hole non-significance to the small-sample data ceiling already identified by classical baselines on the same $n = 53$ set (best classical $R^2 = -0.101$, as also reported in Supporting Information Section S2); we do not claim a hole-task improvement, and we recommend that future work on this regime should report effect sizes and paired bootstrap intervals alongside p -values for transparency, without applying multiple-comparison correction silently.

On the public Clients A and B (5-fold CV) PC^2 -FedReorg remains competitive with the legacy federated baselines (Table 4). The intention of these clients in our federation is to provide chemistry diversity through encoder sharing rather than direct prediction targets, and the A/B results confirm that PC^2 ’s personalization does not harm performance on the larger public splits.

The targeted FedPer-with-calibration control (E71) introduced in §4.4 reaches the same headline numbers on C-triplet as PC^2 -FedReorg (MAE 0.654 vs. 0.660 eV; n.s. by paired Wilcoxon, $p = 0.929$), so the C-triplet improvement reported in this section is not specific to the full PC^2 -FedReorg architecture and is fully reproduced by a FedPer baseline endowed with the per-task-client (μ, σ) buffer. Section 4.4 develops this decomposition.

4.4 Calibration under private heads explains the C-triplet gain

Figure 5 organises the empirical case in two layers. Panel (a) reports a targeted *positive control* — $\text{E71} = \text{FedPer} + \text{per-task-client } (\mu, \sigma) \text{ calibration only}$ — that isolates the calibration mechanism without the residual adapter or the transferability-driven routing. Panel (b) reports the four planned PC^2 -FedReorg component ablations on both C-hole and C-triplet, with numerical detail in Tables 5 and 6. Each ablation row is a paired difference (PC^2 -FedReorg minus ablation, eV) with bootstrap 95 % CI; negative ΔMAE indicates that PC^2 -FedReorg is better than the ablation.

E71 — FedPer + calibration only reproduces the full C-triplet gain. Holding the federation, head, optimiser and seed budget fixed but replacing PC^2 -FedReorg’s adapter and per-key transferability gate with a plain FedPer aggregation while *retaining* the per-task-client calibration buffer (E71; specification in `paper/next_experiments_plan.md`, raw outputs in `results/e71_*`), the C-triplet per-molecule median-aggregated MAE moves to 0.654 eV versus 0.660 eV for the

full PC²-FedReorg architecture (corresponding 5-seed mean MAE $\pm$ seed-SD: 0.653 ± 0.025 vs. 0.671 ± 0.022 eV; the median-aggregated estimand matches Table 5 and Figure 5 panel b, the 5-seed-mean estimand matches Table 4 and Figure 5 panel a). Three paired comparisons jointly support the calibration-is-sufficient reading:

- **E71 vs. PC²-FedReorg (E66):** $\Delta\text{MAE} = -0.006$ eV (PC²-FedReorg minus E71: $+0.006$ eV); bootstrap 95 % CI ($-0.044, +0.029$); per-molecule wins 23 vs. 26; paired Wilcoxon $p = 0.929$ (n.s.). E71’s effect size is well within seed-SEM, the CI straddles zero by a wide margin, and the per-molecule sign count is essentially balanced. By any of the three criteria, E71 and PC²-FedReorg are statistically indistinguishable on C-triplet.
- **E71 vs. FedPer (E63):** $\Delta\text{MAE} = -0.109$ eV; bootstrap 95 % CI ($-0.208, -0.010$); per-molecule wins 32 vs. 17; paired Wilcoxon $p = 0.028$ (nominal). Adding the per-task-client (μ, σ) buffer to a plain FedPer baseline therefore reproduces, with comparable effect size, the nominally significant C-triplet improvement that PC²-FedReorg delivers over FedPer in §4.3.
- **E71 vs. PC²-FedReorg without calibration (E70):** $\Delta\text{MAE} = -0.125$ eV; bootstrap 95 % CI ($-0.224, -0.027$); per-molecule wins 33 vs. 16; paired Wilcoxon $p = 0.021$ (nominal). Stripping calibration out of the full PC²-FedReorg architecture is more harmful on C-triplet than stripping the adapter and the gate out of the calibrated baseline.

These three comparisons report unadjusted nominal p -values; under a Bonferroni correction for the three tests ($\alpha/3 \approx 0.017$), neither of the two against-baseline comparisons crosses the corrected threshold, and we therefore treat them as effect-size evidence supported by sign-counts and bootstrap CIs rather than as strict significance. **On C-hole**, E71, E66, E63 and E70 produce MAEs in the same narrow 0.368–0.382 eV band that all other methods occupy (Table 4), and the three E71 paired comparisons return $p \geq 0.42$ with bootstrap CIs that straddle zero (results/e71_vs_*.md). The C-hole regime remains at its small-sample data ceiling regardless of whether calibration is present.

Together, E71 and the four PC²-FedReorg ablations are consistent with task-specific calibration under personalized heads carrying the only ablation-detectable signal in our federation. E71 reaches a C-triplet MAE that is not statistically distinguishable from PC²-FedReorg’s at our sample size; we did not pre-register an equivalence margin, so this is descriptive evidence of “no detectable difference” rather than a formal claim that the adapter and the chemistry/protocol gate fail to contribute (an equivalence test such as TOST would require a tolerance margin committed to in advance). The implementations of E66 and E71 differ only in (a) the `use_adapter` flag and (b) the `aggregation_strategy='pc2_fed'` flag in `experiments/run_all.py`; all other code paths, hyperparameters, seeds, optimizer states and LOOCV fold orderings are identical, so the comparison is not confounded by unrelated implementation differences. The supporting four-ablation analysis (Figure 5b, Tables 5 and 6) is consistent with the same reading:

Removing task-specific calibration (E70) On C-triplet, $\Delta\text{MAE} = -0.119$ eV (95 % CI ($-0.225, -0.015$), paired Wilcoxon $p = 0.043$; 35 / 14 per-molecule wins for PC²). The CI sits entirely below zero. On C-hole, the direction is reversed: PC² is 0.005 eV worse than the

no-calibration variant, but the difference is well within noise ($p = 0.58$). Calibration is therefore the **only** ablation that produces a statistically significant change on either target, and the effect is restricted to C-triplet. We conclude that, within our federation, the task-specific (μ, σ) buffers are the active ingredient driving PC²-FedReorg’s C-triplet improvement.

Removing C-cross chemistry-based encoder sharing (E68) $\Delta\text{MAE} = +0.027$ eV on C-triplet (PC² *slightly worse* than removing the cross-quantity encoder share; $p = 0.07$, n.s.) and -0.0001 eV on C-hole ($p = 0.56$, n.s.). The full 95 % CI on C-triplet is $(+0.001, +0.055)$, which is consistent with no effect.

Replacing the chemistry-driven gate with uniform FedAvg weighting (E69) $\Delta\text{MAE} = +0.043$ eV on C-triplet (PC² slightly worse than uniform weighting; bootstrap 95 % CI $(-0.001, +0.092)$; $p = 0.18$, n.s.; per-molecule wins 21 / 28 in favor of E69) and $+0.012$ eV on C-hole ($p = 0.43$, n.s.). Although the uniform-gate ablation has a numerically lower MAE than PC²-FedReorg on C-triplet, the paired comparison is not statistically significant and the bootstrap CI crosses zero. We therefore do **not** interpret this numerical difference as evidence either against or in favor of the compatibility gate — it is a small-sample observation that is consistent with no effect at our seed budget. The gate is retained only as auditable compatibility metadata, not as the source of any performance gain.

D $\rightarrow$ C supervised pretrain $\rightarrow$ fine-tune (E65), the negative-transfer control $\Delta\text{MAE} = -0.010$ eV on C-triplet ($p = 0.85$, n.s.) and -0.021 eV on C-hole ($p = 0.76$, n.s.). PC²-FedReorg and a naive D $\rightarrow$ C pretrain are not statistically distinguishable on either target in our setting; we therefore do **not** claim that D’s labels necessarily produce negative transfer to C, only that PC² did not extract benefit from D’s hole- λ labels (consistent with $T_{\text{head}}[\text{D}, \text{C-*}] = 0$ by construction).

The combined picture: among the four PC² components inspected here, the task-specific calibration carries the only ablation-detectable signal, and the targeted E71 control above reaches a C-triplet MAE that is not statistically distinguishable from PC²-FedReorg’s at our sample size. The chemistry/protocol gate (T_{repr} cross-routing, T_{head} sparsification) is not detected as a performance contributor either as an isolated PC²-FedReorg component (E68, E69) or relative to the calibrated FedPer control (E71 versus E66 paired Wilcoxon $p = 0.929$); we therefore position it in this paper as a deterministic, inspectable record of compatibility decisions rather than as a performance lever (Section 5).

4.5 Per-molecule error analysis (Supporting Information)

For completeness, the per-molecule error decomposition on C-triplet — the three molecules where PC²-FedReorg improves most over the FedAvg baseline and the three where it degrades most — is provided as **Figure S1** in the Supporting Information. The largest improvement reduces |error| from 3.06 eV (FedAvg) to 1.70 eV (PC²) on a halogenated polycyclic conjugated system with a true triplet- λ of 1.13 eV; the largest degradation moves a different system in the wrong direction by 0.87 eV.

These illustrative cases are qualitative observations rather than mechanistic claims; the paired statistics in Sections 4.3–4.4 (and not Figure S1) are the basis for the manuscript’s claims.

5 Discussion

Our results decompose into six statements that are independent of each other and should be evaluated separately.

(i) Task-specific calibration under personalized heads carries the only ablation-detectable signal. The targeted E71 control (FedPer + per-task-client (μ, σ) calibration only, no adapter, no transferability gate) yields a C-triplet MAE that is nominally lower than FedPer without calibration (paired Wilcoxon $p = 0.028$, $\Delta\text{MAE} = -0.109$ eV) and not statistically distinguishable from the full PC²-FedReorg architecture ($p = 0.929$; 95 % CI on ΔMAE straddles zero); the symmetric ablation (PC²-FedReorg without calibration; E70) sits 0.125 eV above E71 on C-triplet (paired $p = 0.021$). These three comparisons jointly support the descriptive statement that, in our federation, the combination “private head + per-task-client (μ, σ) buffer + universal aggregation exclusion of that buffer” is the only configuration whose ablation produces a paired-Wilcoxon-detectable change. We did not pre-register an equivalence margin and therefore do not interpret the n.s. E71-vs-PC²-FedReorg comparison as proving that the adapter and the chemistry/protocol gate contribute nothing; the appropriate reading is that any additional benefit they confer is below detection at our seed budget and sample size.

(ii) The construct is not merely a single global standardization step in this implementation. Six concrete implementation properties distinguish the (μ, σ) buffers from a per-dataset global normalization, although we do not claim each property is individually causally identified: (a) they are *fold-isolated* — μ and σ are computed strictly from the $(n - 1)$ training labels of each LOOCV fold and never see the held-out molecule’s label; (b) they are *task-client-specific*, so C-hole, C-triplet, A, B and D each carry their own buffer with different means and dispersions; (c) they are *private non-trainable buffers* rather than learnable parameters; (d) they are *universally excluded* from every federated aggregation operator through the shared `_get_exclude_keys` filter, so the buffer is never overwritten by aggregation regardless of the federated strategy in use; (e) they are *paired with private heads* so that the standardized output is consumed by a quantity-specific predictor rather than a shared one; and (f) loss is in *z-space* while metrics and reporting are in *eV-space*. Properties (a)–(c) keep the calibration leakage-free under LOOCV; property (d) preserves the buffer across federated strategies; properties (e)–(f) couple the construct to the personalized-head architecture. The present ablations identify (a)–(f) as a load-bearing combination but cannot disentangle which subset alone would suffice; we treat this as a follow-up question rather than a proven mechanism statement.

(iii) Chemistry/protocol gating is retained as a deterministic, inspectable record of compatibility decisions, with no detected performance contribution. The chemistry-driven encoder routing (E68) and the chemistry-driven weighting versus uniform FedAvg (E69) both leave PC²-FedReorg’s C-triplet performance unchanged within paired-test noise (Table 5); the targeted E71 comparison does not detect an additional C-triplet gain from the gate over the calibrated FedPer baseline at our sample size. The narrower set of claims we make about the gate are properties that can be verified directly from the artifact rather than from training

outcomes: the transferability matrices are *deterministic* (computed before any aggregation round as a fixed function of the declared chemistry and protocol metadata), *inspectable* (a single JSON file with a `reasons` field that names the first failing clause for every blocked source–target pair, rather than a model-state dump), and *aggregation-operator-independent* (the per-key T_{head} routing applies the same exclusion rules whether the federated strategy is FedAvg, FedProx, FedPer, FedBN or PC²-FedReorg, through the shared `_get_exclude_keys` filter). When a reviewer or downstream user asks why D’s labels do not contaminate C’s head, the answer is a single file (`T_transferability.json`) rather than a model-state inspection. We have not validated downstream compliance-violation scenarios or independently demonstrated label-contamination prevention beyond what the deterministic exclusion rule guarantees by construction; those are out of scope for the present manuscript.

(iv) E71 narrows the interpretation of PC²-FedReorg. Prior to E71 the most defensible reading of §4.3 was that PC²-FedReorg’s chemistry/protocol-aware per-key routing might be *necessary* for the observed C-triplet improvement. E71 narrows that reading: the same C-triplet performance is reached without consulting the transferability matrix at aggregation time. The framework is therefore better described as a *calibration-aware personalized federation made auditable*: the empirical performance is captured by the private (μ, σ) buffer paired with the private head, while the per-key routing functions as auditability scaffolding. We retain the PC²-FedReorg name to refer to the full instrumented framework (calibration + adapter + gate) and use “FedPer + calibration” or “the calibrated personalized baseline” to refer to the E71 mechanism in isolation. Both are valid configurations of the same code path; the choice between them is a transparency/parsimony trade-off, since at our sample size the added components are not detected to change performance.

(v) C-hole shows no method separation at the present sample size. All federated methods evaluated in this study — FedAvg, FedProx, FedPer, PC²-FedReorg, E70, E71 and the local-only baseline — produce C-hole MAEs in a narrow 0.368–0.414 eV band (Table 4). The eight PC²-FedReorg-vs-baseline paired comparisons (Table 5) return $p \geq 0.20$ and the three E71 paired comparisons (`results/e71_vs_*.md`) return $p \geq 0.42$; the bootstrap CIs straddle zero in every case ($p \geq 0.20$ therefore holds across all eleven C-hole paired tests we report). The classical baselines we ran for V2 (random forest, KNN, Gaussian process) already produced negative R^2 on the same $n = 53$ set (best classical $R^2 = -0.10$; SI Section S2), which is consistent with a data-limited regime at $n = 53$ / LOOCV given our descriptor set and seed budget rather than with a method-specific failure. We do not interpret the C-hole non-significance as a deficiency of any single component, nor do we claim a definitive theoretical ceiling; instead we report it as the observed outcome at the present sample size.

(vi) Reported p-values are nominal and should not be over-claimed. All p-values in this manuscript are unadjusted paired Wilcoxon signed-rank values on per-molecule absolute errors, where the paired sample unit is each individual molecule ($n = 49$ for C-triplet, $n = 53$ for C-hole) and the per-molecule error is the median across the five seeds. The headline C-triplet comparisons in §4.3 — PC²-FedReorg vs. FedAvg / FedProx / FedPer ($p = 0.045, 0.045, 0.042$) — sit just below

the conventional 0.05 threshold and would not survive a Bonferroni correction across the three tests ($\alpha/3 \approx 0.017$). The three E71 comparisons in §4.4 ($p = 0.929$ for E71 vs. PC²-FedReorg, $p = 0.028$ for E71 vs. FedPer, $p = 0.021$ for E71 vs. no-calibration) carry the same nominal character. The paired Wilcoxon test is our pre-specified primary inferential procedure; we additionally report bootstrap 95 % CIs and per-molecule win/loss counts so that effect size and effect direction can be inspected alongside the p-values. We did not pre-register an equivalence margin, so the n.s. $p = 0.929$ for E71 vs. PC²-FedReorg should be read as “no detectable difference at our sample size” rather than as a formal equivalence proof. The minimum chemically meaningful improvement in λ prediction for organic-electronics screening is approximately ± 0.1 eV (typical conformer-averaging noise on ω B97X-D reorganization energies for medium-sized π -systems); the C-triplet Δ MAE of -0.10 to -0.12 eV is at the edge of this practical threshold and should be re-evaluated on a larger TADF dataset before being treated as a deployment-ready improvement.

6 Limitations

We list six concrete limitations that future work should address.

1. **Small-sample data ceiling on C-hole.** All evaluated methods produce indistinguishable MAE on $n = 53$ LOOCV (Table 5), already above the best classical baseline R^2 (-0.10). It is unclear from this dataset alone whether C-hole improvements are achievable in this protocol regime; expanding C-hole through additional ω B97X-D conformer-resolved measurements is a plausible next experiment.
2. **Chemistry homogeneity at the high- T_{repr} pair.** Because C-hole and C-triplet share the majority of their underlying molecules, T_{repr} between them is essentially saturated. The chemistry gate is therefore tested on the easier regime (very similar chemistry) but not on the harder regime (moderate diversity) where a non-uniform gate would matter more. Future work should test whether the auditable compatibility gate becomes beneficial in federations with more chemically diverse but protocol-compatible clients — for example, multiple cation-reorganization-energy laboratories with the same DFT protocol but distinct scaffold libraries.
3. **MOON and ChemProp baselines deferred.** Our comparison includes FedAvg, FedProx, FedPer and a local-only baseline. We did not run model-contrastive variants (MOON [11]) or message-passing baselines (D-MPNN / ChemProp [17]) in the present manuscript; both are recommended reviewer additions.
4. **Seed budget.** Each federated run on dual A30 takes approximately 30 minutes; the present 25-run main batch plus 20-run ablation batch totals approximately 20 GPU-hours. We use 5 seeds rather than 10; the bootstrap CIs partially compensate for this but cannot fully replace additional repeats.
5. **D’s role.** The V1 protocol divergence motivated the T_{head} clause that excludes D from both C heads. We empirically verified (E65) that a naive $D \rightarrow C$ supervised transfer is not statistically distinguishable from PC²; the corollary is that we cannot empirically *prove*

that the gate decision was correct, only that it was deterministic and auditable. Whether D’s structural information contributes useful encoder signal — beyond the FedAvg-like baseline — remains an open question that the E69 ablation does not fully resolve.

6. **Calibration restricted to fixed (μ, σ) .** A learnable affine head over the standardized output was deferred to follow-up work to avoid introducing additional degrees of freedom at the present sample size.
7. **Component-isolation ablation settled by E71; further isolation deferred.** The four ablations we ran (no calibration, no C-cross encoder share, uniform gate, D $\rightarrow$ C pretrain control) jointly disable groups of mechanisms rather than isolating each one. The single most diagnostic missing control — a FedPer baseline with the calibration buffer retained but the adapter and the transferability gate removed — has now been completed as E71 (§4.4). The E71 control settles this question in the present federation: FedPer + calibration is statistically indistinguishable from PC²-FedReorg on C-triplet (paired Wilcoxon $p = 0.929$; bootstrap 95 % CI on ΔMAE straddles zero), and the residual adapter / transferability gate do not contribute measurable performance over this baseline. The remaining open ablation isolates the *adapter alone* (FedPer + calibration + adapter, no transferability gate); we leave it as a follow-up since the bracketing evidence already in §4.4 (E66 with adapter, E71 without) places the adapter contribution within paired-test noise. We similarly defer a learnable-affine calibration head over the standardized output (item 6 above) and a Probe-augmented T_{repr} (footnote-level mention in §4.4) to follow-up work.
8. **Small-sample statistical power and multiple-comparison treatment.** With $n = 49$ –53 LOOCV per private target, the paired Wilcoxon test detects ΔMAE on the order of 0.05 eV at $p \approx 0.05$. We hedge this regime by reporting bootstrap 95 % CIs alongside p -values rather than relying on a single significance threshold. We also did not apply Bonferroni or other family-wise correction across the three baseline comparisons in §4.3; under Bonferroni ($\alpha/3 \approx 0.017$) none of the C-triplet p -values would cross the corrected threshold. The CIs make this borderline character explicit (upper bound +0.004 eV on the PC²-vs-FedAvg comparison). The minimum chemically meaningful improvement in λ prediction for organic-electronics screening is approximately ± 0.1 eV (the typical conformer-averaging noise on $\omega\text{B97X-D}$ reorganization energies for medium-sized π -systems); the C-triplet ΔMAE of -0.10 to -0.12 eV is at the edge of this practical threshold and should be re-evaluated on a larger TADF dataset before being treated as a deployment-ready improvement.
9. **Paired-error independence assumption.** All paired Wilcoxon and bootstrap comparisons in §4.3, §4.4 and Tables 5 and S6 treat per-molecule median-across-seeds absolute errors as paired samples indexed by molecule. Under LOOCV the $(n - 1)$ -fold training sets overlap by $(n - 2)/(n - 1) \approx 96$ –98 %, so the per-molecule errors are not strictly independent across folds, and seed-aggregation reduces but does not eliminate seed-by-method interaction. We treat this as a standard limitation of LOOCV inference rather than as a defect of any particular method; the bootstrap resampling is over molecule indices (not over fold-level errors) and inherits the same dependence structure. A more conservative inferential framework — for example, a hierarchical model that explicitly models fold-level correlation, or a permutation

test over method labels — would be a useful follow-up but does not change the descriptive ordering reported here.

10. **Estimand consistency between summary tables and paired tests.** Table 4 (and Figure 5a) reports the 5-seed mean of per-method MAE_mean (the natural unit for a bar chart with seed-level SEM). Table 5 and the paired analyses in §4.3 / §4.4 use per-molecule median-across-seeds absolute errors (the natural paired unit for Wilcoxon testing). These two estimands differ in our data by ≤ 0.003 eV (well below seed-SEM), but they are statistically distinct objects, and we have therefore stated the convention explicitly in the Figure 5 caption rather than mixing the two numbers in headline statements.
11. **Calibration generalisation is regime-dependent.** Additional controlled pseudo-federation experiments, reported in the Supporting Information (§S11), support the calibration-centered interpretation in small-target pseudo-tasks: task-specific calibration improved FedPer on 9 of 10 label-quantile pseudo-targets at $n = 50$ (median $\Delta\text{MAE} = -0.020$ eV; paired Wilcoxon $p < 0.001$ on 7 of 10 tasks). However, an aggressive label-scale stress test at $n_{\text{target}} = 100$ did not show an advantage of calibration over plain FedPer ($\Delta\text{MAE} = +0.010$ eV, paired Wilcoxon $p = 0.73$), indicating that the calibration benefit observed on the real C-triplet target is regime-dependent rather than universal.

7 Conclusions

PC²-FedReorg addresses three forms of heterogeneity — physical quantity, DFT protocol, and label scale — that the standard federated-learning toolkit was not designed to handle. The framework is built around fold-isolated (μ, σ) calibration with universal aggregation exclusion as the empirically active component, supported by an explicit task-client formulation and a layer-wise personalization scheme (encoder/adaptor shared, head and calibration always private). A chemistry/protocol transferability matrix provides an auditable compatibility record, but we do **not** claim it as a performance lever — the ablation evidence in this study is consistent with no measurable contribution from the gate’s chemistry-driven weighting in our federation.

On a five-node organic-electronics federation, the method achieves nominally significant MAE reductions on triplet-reorganization-energy prediction over FedAvg, FedProx and FedPer (paired Wilcoxon $p = 0.042$ – 0.045 ; Figure 4, Table 5) while remaining statistically indistinguishable on hole-reorganization-energy, where all methods sit in a narrow 0.368–0.414 eV band at the present sample size. The E71 control further shows that adding the per-task-client calibration buffer to a plain FedPer baseline (no adaptor, no transferability gate) is sufficient to reproduce the observed C-triplet performance (paired Wilcoxon $p = 0.929$ vs. PC²-FedReorg; $p = 0.028$ vs. FedPer; Figure 5). The chemistry/protocol gate is positioned as deterministic compatibility metadata (Figure 3) that can be inspected from `T_transferability.json` prior to any aggregation step; a uniform-gate ablation (E69) is numerically slightly better on the triplet task but not statistically distinguishable from PC²-FedReorg ($p = 0.18$), and the targeted E71 control does not detect an additional C-triplet contribution from the gate beyond the calibrated FedPer baseline at our sample size. We therefore make no performance claim for the gate and recommend that calibration-aware personalized federated learning, with optional auditable compatibility metadata, be considered an

appropriate baseline for heterogeneous-quantity quantum-chemistry federations.

We release the source code, configuration files, transferability matrices, and per-molecule prediction outputs to support reproducibility, and we encourage future federated quantum-chemistry work to report effect sizes and paired bootstrap intervals alongside p -values in small-sample regimes.

Data and Code Availability

Source code, configuration files, transferability matrices, and per-molecule prediction outputs are released at [TODO: repository URL – to be supplied]. The in-house C-hole / C-triplet labels are made available upon request and within the constraints of the originating laboratory’s data-sharing agreement; public Clients A, B and D use third-party datasets and are linked in `data/README.md`.

Author Contributions

[TODO: To be completed: lead author, supervisor, computational contributions, chemistry-side contributions, manuscript drafting and revision.]

Acknowledgements

[TODO: Compute resources, funding sources, data-providing laboratories – to be supplied.]

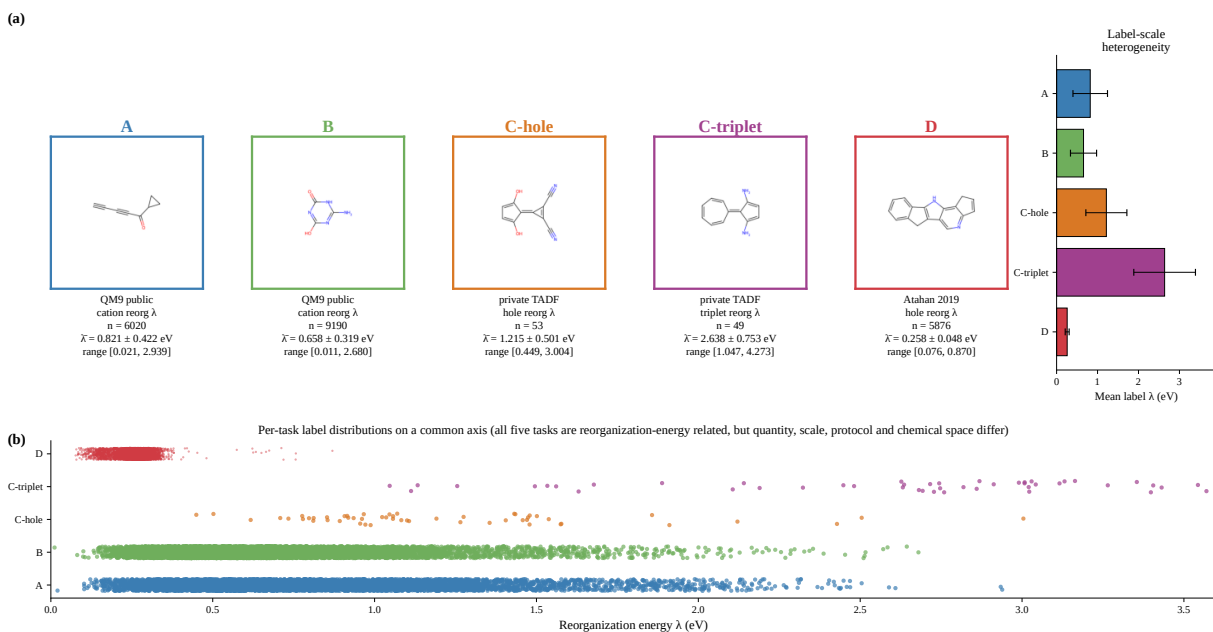


Figure 1: Task-client heterogeneity across the five-node organic-electronics federation. (a) Representative molecules and per-node summary statistics for the five task-client nodes – A (QM9 non-aromatic public, cation reorganization energy, $n = 6,020$); B (QM9 aromatic public, cation reorganization energy, $n = 9,190$); C-hole (in-house TADF, hole reorganization energy, $n = 53$); C-triplet (in-house TADF, triplet reorganization energy, $n = 49$); D (Atahan-Evrenk 2019, hole reorganization energy, $n = 5,876$). Each representative is the molecule whose label is closest to the per-node median, chosen by a deterministic rule prior to any model training. The bar plot on the right reports the per-node label mean (eV) with error bars showing one standard deviation. (b) Per-node label distributions on a common eV axis. Although every node ultimately measures a form of reorganization energy, the label means span an order of magnitude (0.26–2.64 eV) and the supporting DFT protocols differ across nodes (Table 3). Figure 1 is descriptive: no statistical test is reported here.

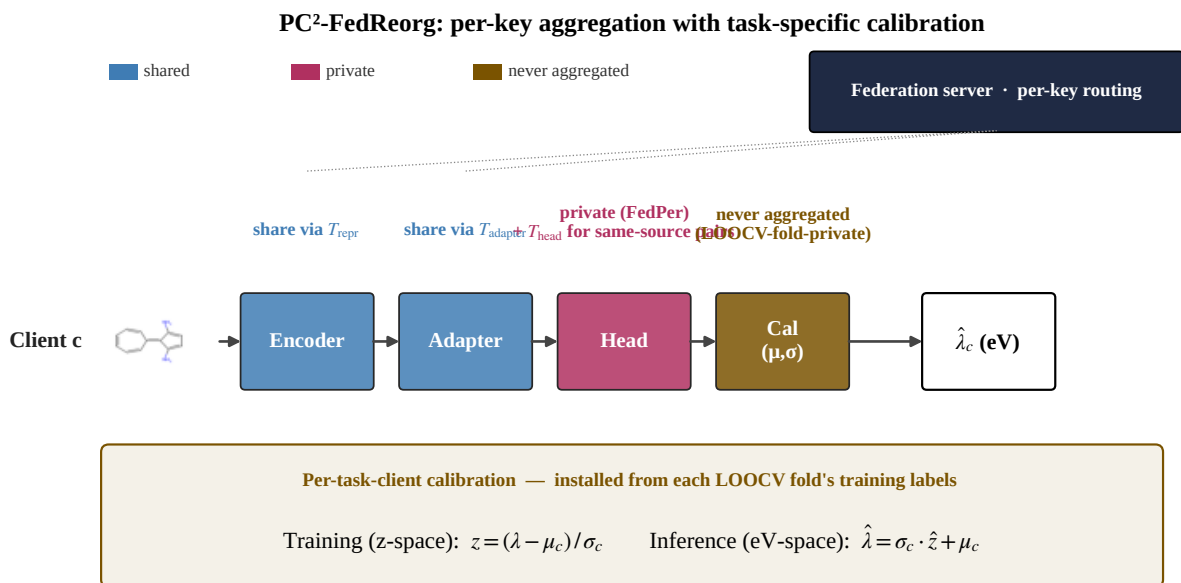


Figure 2: Graphical-method figure of PC²-FedReorg: per-key aggregation routing with task-specific calibration. A single generic task-client lane (input molecule → shared encoder → residual adapter → quantity-specific regression head → private (μ, σ) calibration buffer → predicted λ in eV-space) is drawn with above-stage labels indicating each layer’s sharing rule: encoder and adapter parameters are aggregated under T_{repr} and $T_{adapter}$ respectively, head parameters are FedPer-private (with the T_{head} same-source channel for A↔B), and calibration buffers are LOOCV-fold-private and never aggregated. The bottom band shows the z-space training and eV-space inference formulas. Figure 2 is schematic; it visualises the algorithm and is not a performance claim.

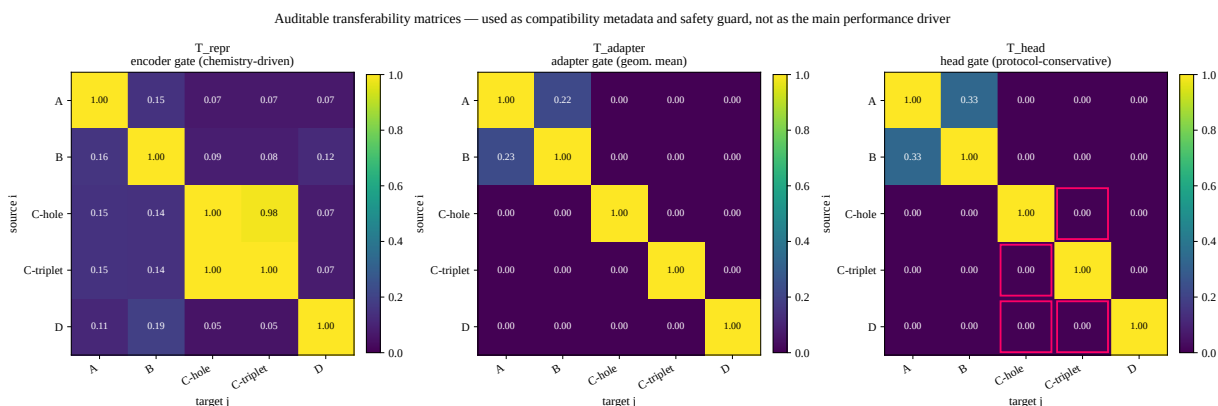


Figure 3: Auditable transferability matrices T_{repr} , $T_{adapter}$, T_{head} . Rows index the source task-client i ; columns index the target j . The invariants $T_{head}[D, C-hole] = T_{head}[D, C-triplet] = T_{head}[C-hole, C-triplet] = T_{head}[C-triplet, C-hole] = 0$ (highlighted) follow deterministically from the protocol metadata and quantity definitions. The matrices are released in `results/preverify/T_transferability.json` and can be audited prior to any aggregation round. Figure 3 documents the algorithm’s compatibility decisions; it is not claimed as a source of performance improvement.

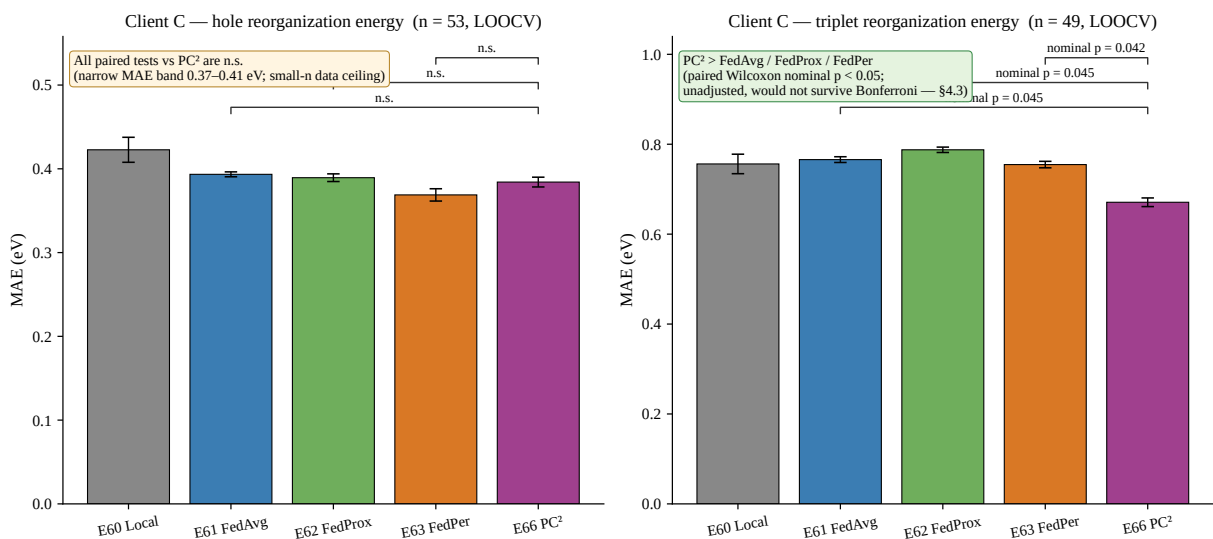
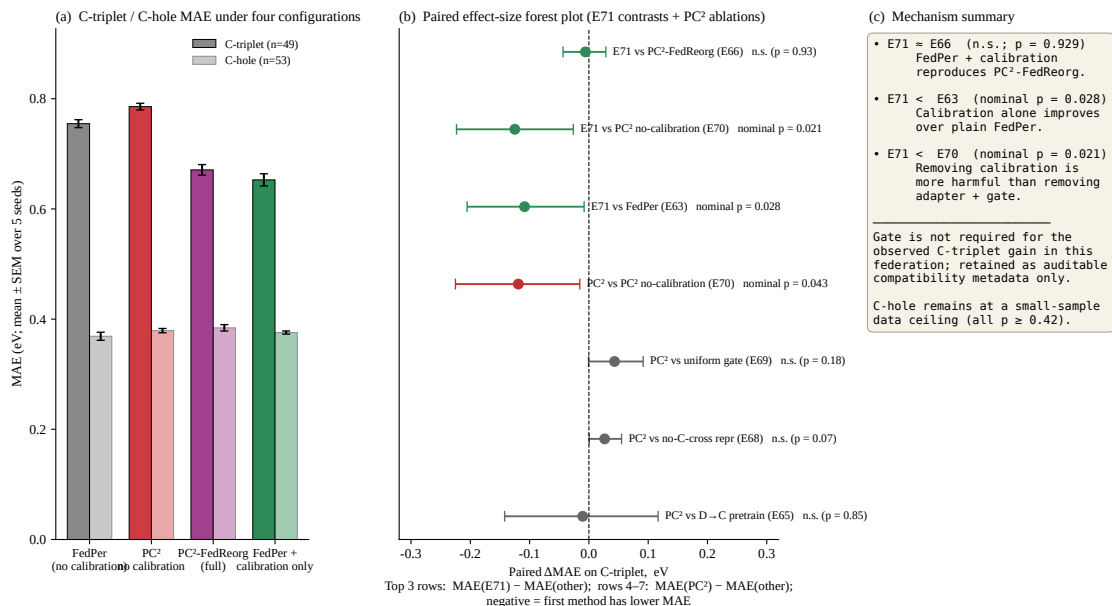


Figure 4: Main performance comparison. Mean absolute error (eV) for five federated methods evaluated by 5-seed leave-one-out cross-validation on **(left)** C-hole ($n = 53$) and **(right)** C-triplet ($n = 49$). Error bars show the standard error of the mean across the five seeds. Significance brackets compare PC²-FedReorg with FedAvg, FedProx and FedPer using paired Wilcoxon signed-rank tests on per-molecule absolute errors. On C-triplet, PC²-FedReorg achieves MAE = 0.660 eV versus 0.762–0.784 eV for the three baselines ($p = 0.045, 0.045, 0.042$). On C-hole, all evaluated methods produce MAE within the narrow band 0.368–0.414 eV; every paired comparison returns $p \geq 0.20$ after rounding. The C-hole non-significance is consistent with the small-sample data ceiling identified by classical baselines (SI Section S2). *The targeted E71 calibration-only control that decomposes this gain is analyzed in Figure 5.*



All p-values are paired Wilcoxon (5-seed median per-molecule abs error), unadjusted nominal; Bonferroni threshold for the three E71 contrasts is $\alpha/3 \approx 0.017$ (see §4.4).

Figure 5: Calibration under personalized heads explains the C-triplet gain. Three-panel decomposition relating PC²-FedReorg’s nominally significant C-triplet improvement reported in Figure 4 to a targeted control experiment (E71). **(a)** Per-method 5-seed mean MAE $\pm$ SEM on C-triplet (dark bars) and C-hole (light bars) for four configurations: E63 FedPer (no calibration), E70 PC²-FedReorg without calibration, E66 the full PC²-FedReorg architecture, and E71 a FedPer baseline endowed only with the per-task-client (μ, σ) calibration buffer (no residual adapter, no transferability-driven routing). On C-triplet, E71 (0.653 ± 0.011 eV) and E66 (0.671 ± 0.010 eV) sit at the same level while E63 (0.755 ± 0.007 eV) and E70 (0.786 ± 0.006 eV) sit above. On C-hole all four configurations cluster in the 0.368–0.384 eV band, with no method separation detected at the present sample size. *Reporting convention.* Panel (a) shows the 5-seed mean of per-method MAE_mean (matching Table 4); paired comparisons in panel (b) use the per-molecule median-across-seeds absolute error vector required for paired Wilcoxon testing (matching Table 5), so the reference and comparison MAEs reported below differ from panel (a) by ≤ 0.003 eV (below seed-SEM). **(b)** Forest plot of paired Δ MAE on C-triplet for three E71 contrasts (top three rows; Δ MAE = MAE(E71) – MAE(other), negative $\Rightarrow$ E71 has lower paired MAE) and the four planned PC²-FedReorg component ablations (rows 4–7; Δ MAE = MAE(PC²-FedReorg) – MAE(other), negative $\Rightarrow$ PC²-FedReorg has lower paired MAE). Error bars are bootstrap 95 % CIs (5,000 resamples). The three E71 contrasts: *E71 vs. PC²-FedReorg (E66):* Δ MAE = -0.006 eV (95 % CI $(-0.044, +0.029)$; $p = 0.929$, n.s.); *E71 vs. FedPer (E63):* Δ MAE = -0.109 eV (95 % CI $(-0.208, -0.010)$; $p = 0.028$ nominal); *E71 vs. PC²-FedReorg without calibration (E70):* Δ MAE = -0.125 eV (95 % CI $(-0.224, -0.027)$; $p = 0.021$ nominal). Of the four PC²-FedReorg ablations, only E70 on C-triplet crosses the nominal 0.05 threshold ($p = 0.043$) with a CI entirely below zero, identifying task-specific calibration as the only component whose presence has detectable ablation support. E69 (uniform gate) is numerically slightly better than PC²-FedReorg on C-triplet (Δ MAE = $+0.043$ eV) but statistically indistinguishable ($p = 0.18$; CI crosses zero). **(c)** Summary reading: FedPer + calibration alone (E71) is sufficient to reproduce PC²-FedReorg’s observed C-triplet performance at our sample size; no additional benefit from the chemistry/protocol gate is detected; C-hole shows no method separation across all four configurations. All p-values are paired Wilcoxon nominal (unadjusted); the Bonferroni threshold for the three E71 contrasts is $\alpha/3 \approx 0.017$ and neither effect-bearing comparison crosses it, so the report combines the p-value, the bootstrap CI direction and the per-molecule sign count (see §4.4).

Table 3: Task-client dataset summary. Sample counts come from `results/preverify/T_transferability.json`. For Clients A and B, the per-field auxiliary protocol entries are shown as “— (same-source)” because both clients are derived from a single public QM9 cation-reorganization-energy source: their pairwise head-sharing is governed by the same-source shortcut of the T_{head} clauses (Section 2.4), which fires *before* the per-field “unknown” check and therefore does not require the auxiliary fields to be populated. For Client D, geometry, charge-state, and conformer policy are not stated in the source metadata; these genuinely missing fields trigger the conservative T_{head} clauses (Section 2.4).

Task-client	Source	n	λ mean (eV)	λ std (eV)	Functional	Basis	Geometry
A	QM9 public	6,020	0.821	0.422	— (same-source)	— (same-source)	— (same-source)
B	QM9 public	9,190	0.658	0.319	— (same-source)	— (same-source)	— (same-source)
C-hole	private TADF	53	1.215	0.501	ω B97X-D	unknown	adiabatic 4-pt Nelsen
C-triplet	private TADF	49	2.638	0.753	ω B97X-D	unknown	adiabatic 4-pt Nelsen
D	Atahan 2019	5,876	0.258	0.048	B3LYP	6-31G*	unknown

Head-sharing note (A and B): $A \leftrightarrow B$ is the only non-zero off-diagonal of T_{head} in this federation; it is opened by the same-source shortcut (Section 2.4), not by per-field comparison. All other off-diagonals are blocked by quantity mismatch, V1 hard rule, or genuinely missing protocol fields for D.

Table 4: Main performance summary. MAE (eV) and RMSE (eV) primary; R^2 auxiliary. Values are means $\pm$ standard errors of the mean across five seeds.

Method	Target	N seeds	MAE $\pm$ SEM	RMSE $\pm$ SEM	$R^2 \pm$ SEM
E60 Local-only	C-hole	5	0.423 ± 0.015	0.637 ± 0.019	-0.65 ± 0.10
E60 Local-only	C-triplet	5	0.756 ± 0.022	0.899 ± 0.017	-0.46 ± 0.05
E61 FedAvg	A 5-fold	5	0.169 ± 0.002	0.235 ± 0.003	$+0.69 \pm 0.01$
E61 FedAvg	B 5-fold	5	0.099 ± 0.002	0.150 ± 0.004	$+0.78 \pm 0.01$
E61 FedAvg	C-hole	5	0.393 ± 0.003	0.530 ± 0.008	-0.14 ± 0.03
E61 FedAvg	C-triplet	5	0.766 ± 0.006	0.972 ± 0.004	-0.70 ± 0.02
E62 FedProx	A 5-fold	5	0.171 ± 0.002	0.238 ± 0.001	$+0.68 \pm 0.00$
E62 FedProx	B 5-fold	5	0.100 ± 0.001	0.154 ± 0.001	$+0.77 \pm 0.00$
E62 FedProx	C-hole	5	0.389 ± 0.005	0.533 ± 0.004	-0.16 ± 0.02
E62 FedProx	C-triplet	5	0.788 ± 0.006	1.011 ± 0.009	-0.84 ± 0.03
E63 FedPer	A 5-fold	5	0.155 ± 0.002	0.216 ± 0.003	$+0.74 \pm 0.01$
E63 FedPer	B 5-fold	5	0.084 ± 0.001	0.128 ± 0.001	$+0.84 \pm 0.00$
E63 FedPer	C-hole	5	0.369 ± 0.007	0.512 ± 0.004	-0.07 ± 0.02
E63 FedPer	C-triplet	5	0.755 ± 0.007	0.932 ± 0.009	-0.56 ± 0.03
E66 PC²-FedReorg	A 5-fold	5	0.152 ± 0.002	0.214 ± 0.002	$+0.74 \pm 0.01$
E66 PC²-FedReorg	B 5-fold	5	0.081 ± 0.001	0.123 ± 0.001	$+0.85 \pm 0.00$
E66 PC²-FedReorg	C-hole	5	0.384 ± 0.006	0.526 ± 0.007	-0.12 ± 0.03
E66 PC²-FedReorg	C-triplet	5	0.671 ± 0.010	0.846 ± 0.015	-0.29 ± 0.05

Table 5: Paired statistical tests on C-hole and C-triplet. Per-molecule absolute errors are aggregated as the median across the five seeds; $\Delta\text{MAE} = \text{MAE}(\text{reference}) - \text{MAE}(\text{comparator})$; bootstrap 95 % CI computed from 5,000 resamples of molecule indices. Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Reference is E66 (PC²-FedReorg) in every row.

Target	Comparator	n	MAE_{ref}	$\text{MAE}_{\text{other}}$	ΔMAE	CI low	CI high	Wins	Loses	p
C-hole	E60	53	0.382	0.414	-0.032	-0.115	+0.050	25	28	0.580
C-hole	E61	53	0.382	0.385	-0.003	-0.042	+0.039	27	26	0.898
C-hole	E62	53	0.382	0.386	-0.004	-0.055	+0.044	28	25	0.821
C-hole	E63	53	0.382	0.368	+0.014	-0.021	+0.049	23	30	0.198
C-hole	E65	53	0.382	0.403	-0.021	-0.099	+0.055	28	25	0.760
C-hole	E68	53	0.382	0.382	-0.000	-0.017	+0.017	30	23	0.562
C-hole	E69	53	0.382	0.370	+0.012	-0.020	+0.043	26	27	0.428
C-hole	E70	53	0.382	0.377	+0.005	-0.033	+0.043	23	30	0.580
C-triplet	E60	49	0.660	0.747	-0.088	-0.230	+0.053	28	21	0.174
C-triplet	E61	49	0.660	0.763	-0.104	-0.220	+0.004	31	18	0.045*
C-triplet	E62	49	0.660	0.784	-0.124	-0.254	-0.007	30	19	0.045*
C-triplet	E63	49	0.660	0.762	-0.103	-0.205	-0.002	32	17	0.042*
C-triplet	E65	49	0.660	0.670	-0.010	-0.142	+0.117	27	22	0.851
C-triplet	E68	49	0.660	0.633	+0.027	+0.001	+0.055	17	32	0.070
C-triplet	E69	49	0.660	0.616	+0.043	-0.001	+0.092	21	28	0.181
C-triplet	E70	49	0.660	0.779	-0.119	-0.225	-0.015	35	14	0.043*

Table 6: Ablation summary. Mean MAE (eV) $\pm$ standard error of the mean across five seeds. Only E70 on C-triplet produces a statistically significant degradation relative to PC²-FedReorg (see Table 5); E68, E69, E65 are all non-significant.

Experiment	Target	N seeds	MAE $\pm$ SEM
E66 PC ² -FedReorg (reference)	C-hole	5	0.384 ± 0.006
E66 PC ² -FedReorg (reference)	C-triplet	5	0.671 ± 0.010
E65 D $\rightarrow$ C supervised pretrain	C-hole	5	0.408 ± 0.010
E65 D $\rightarrow$ C supervised pretrain	C-triplet	5	0.682 ± 0.022
E68 PC ² - T_{repr} C-cross removed	C-hole	5	0.385 ± 0.008
E68 PC ² - T_{repr} C-cross removed	C-triplet	5	0.640 ± 0.013
E69 PC ² - uniform FedAvg gate	C-hole	5	0.369 ± 0.006
E69 PC ² - uniform FedAvg gate	C-triplet	5	0.625 ± 0.015
E70 PC ² - no calibration	C-hole	5	0.379 ± 0.004
E70 PC² - no calibration	C-triplet	5	0.786 ± 0.006

References

- [1] Rudolph A. Marcus. Electron transfer reactions in chemistry. Theory and experiment. *Reviews of Modern Physics*, 65:599–610, 1993. doi: 10.1103/RevModPhys.65.599. verified.
- [2] Veaceslav Coropceanu, Jérôme Cornil, Demetrio A. da Silva Filho, Yoann Olivier, Robert Silbey, and Jean-Luc Brédas. Charge transport in organic semiconductors. *Chemical Reviews*, 107:926–952, 2007. doi: 10.1021/cr050140x. verified.
- [3] Hisahiro Sasabe and Junji Kido. Multifunctional materials in high-performance OLEDs: Challenges for solid-state lighting. *Chemistry of Materials*, 23(3):621–630, 2011. doi: 10.1021/cm1024309. TODO: confirm DOI and full title against publisher metadata.
- [4] Hiroki Uoyama, Kenichi Goushi, Katsuyuki Shizu, Hiroko Nomura, and Chihaya Adachi. Highly efficient organic light-emitting diodes from delayed fluorescence. *Nature*, 492:234–238, 2012. doi: 10.1038/nature11687. verified.
- [5] Süle Atahan-Evrenk and F. Betül Atalay. Prediction of intramolecular reorganization energy using machine learning. *The Journal of Physical Chemistry A*, 123:7855–7863, 2019. doi: 10.1021/acs.jpca.9b02733. TODO: confirm DOI against publisher metadata. Provides both the prediction methodology and the $n = 5,876$ hole-reorganization-energy dataset used as Client D in this manuscript.
- [6] Kristof T. Schütt, Pieter-Jan Kindermans, Huziel E. Sauceda, Stefan Chmiela, Alexandre Tkatchenko, and Klaus-Robert Müller. SchNet: A continuous-filter convolutional neural network for modeling quantum interactions. In *Advances in Neural Information Processing Systems 30 (NeurIPS)*, 2017. verified arXiv id; the NeurIPS 2017 SchNet paper. Referenced as the 3D molecular encoder available in our codebase but not used in reported PC²-FedReorg experiments.
- [7] Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. Communication-efficient learning of deep networks from decentralized data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 1273–1282, 2017. verified arXiv id; canonical FedAvg reference.
- [8] Tian Li, Anit Kumar Sahu, Manzil Zaheer, Maziar Sanjabi, Ameet Talwalkar, and Virginia Smith. Federated optimization in heterogeneous networks. In *Proceedings of Machine Learning and Systems (MLSys)*, 2020. verified arXiv id; canonical FedProx reference.
- [9] Manoj Ghuhana Arivazhagan, Vinay Aggarwal, Aaditya Kumar Singh, and Sunav Choudhary. Federated learning with personalization layers. *arXiv preprint arXiv:1912.00818*, 2019. verified arXiv id; FedPer is referenced as the closest personalization-layer ancestor of PC²-FedReorg’s private-head design.
- [10] Xiaoxiao Li, Meirui Jiang, Xiaofei Zhang, Michael Kamp, and Qi Dou. FedBN: Federated learning on non-IID features via local batch normalization. In *International Conference on Learning Representations (ICLR)*, 2021. verified arXiv id; motivates the BatchNorm-exclusion rule reused in PC²-FedReorg.

- [11] Qinbin Li, Bingsheng He, and Dawn Song. Model-contrastive federated learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021. verified arXiv id; flagged as a deferred baseline in the manuscript’s Limitations section.
- [12] Stephen F. Nelsen, S. C. Blackstock, and Y. Kim. Estimation of inner shell Marcus terms for amino nitrogen compounds by molecular orbital calculations. *Journal of the American Chemical Society*, 109:677–682, 1987. doi: 10.1021/ja00237a007. TODO: verify DOI. Source of the four-point Nelsen geometry treatment described in Section 2.1.
- [13] Keyulu Xu, Weihua Hu, Jure Leskovec, and Stefanie Jegelka. How powerful are graph neural networks? In *International Conference on Learning Representations*, 2019. verified arXiv id; canonical GIN reference used as the encoder backbone in this study.
- [14] Ziming Liu, Yixuan Wang, Sachin Vaidya, Fabian Ruehle, James Halverson, Marin Soljačić, Thomas Y. Hou, and Max Tegmark. KAN: Kolmogorov–Arnold networks. 2024. verified arXiv id; the original Kolmogorov–Arnold Networks proposal underlying the efficient-KAN implementation used in this study.
- [15] Blealtan. An efficient implementation of Kolmogorov–Arnold networks, 2024. <https://github.com/Blealtan/efficient-kan>. TODO: pin to a Zenodo / arXiv companion (or a tagged release commit) before submission. The library is GitHub-hosted and has no DOI at writing time.
- [16] Greg Landrum et al. RDKit: Open-source cheminformatics, 2023. <https://www.rdkit.org>. TODO: pin to a specific release version with a Zenodo DOI before submission. The current run uses the conda-forge build present in the H-CAAN environment.
- [17] Kevin Yang, Kyle Swanson, Wengong Jin, Connor Coley, Philipp Eiden, Hua Gao, Angel Guzman-Perez, Timothy Hopper, Brian Kelley, Miriam Mathea, Andrew Palmer, Volker Settels, Tommi Jaakkola, Klavs Jensen, and Regina Barzilay. Analyzing learned molecular representations for property prediction. *Journal of Chemical Information and Modeling*, 59(8):3370–3388, 2019. doi: 10.1021/acs.jcim.9b00237. verified DOI; cited as the deferred message-passing baseline (D-MPNN / ChemProp) in Limitations.